PROJECT DOCUMENTATION

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Introduction

- Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value
- Team Members: List team members and their roles.
- Project Objective: Implement a script-controlled Access Control List (ACL) that restricts record access according to the value of a specified field.

2. Project Overview

- Purpose: Describe the purpose of controlling record-level access dynamically based on field values.
- Scope: Explain the records, users, roles, and field values covered by the access-control solution.
- Key Features: Field-based access restriction, scripted authorization logic, role/user validation, and controlled record visibility.

3. Architecture

- Application Architecture: Describe the application components involved in authentication, authorization, ACL evaluation, and record retrieval.
- ACL Architecture: Explain how the ACL script evaluates the selected field value before allowing or denying access.
- Data Model: Describe the record fields used by the ACL and their relationship to users, roles, or access conditions.

4. Setup Instructions

- Prerequisites: List the required platform, development tools, user roles, and access permissions.
- Configuration: Provide steps to configure the target table/records, ACL, field values, and script logic.
- Environment Setup: Describe any required configuration or environment variables.

5. Folder Structure / Project Structure

- Project Components: Describe the organization of ACL scripts, configuration files, application components, and supporting documentation.
- Script Location: Identify where the access-control script and related configuration are maintained.

6. Running the Application / ACL

- Execution Steps: Explain how to access the target records and trigger the ACL validation.
- Access Flow: Describe the sequence from user authentication to field-value evaluation and access decision.
- Expected Result: Authorized users can access permitted records, while unauthorized users are denied access.

7. ACL / API Documentation

- ACL Definition: Document the ACL type, operation, target table/record, and field used for restriction.
- Script Logic: Explain the conditions checked by the script and the resulting allow/deny decision.
- API or Record Access: Document relevant endpoints or record operations, if applicable, including request details and expected responses.

8. Authentication and Authorization

- Authentication: Explain how users are authenticated before record access is evaluated.
- Authorization: Explain how roles, users, and field values are used to determine whether access is permitted.
- Access Rules: Document the conditions under which a record is allowed or denied.

9. User Interface

- Record View: Provide screenshots showing the record and the field value used for access control.
- Access Behavior: Show the difference between authorized and unauthorized record access.
- Error/Denial Message: Include the message or behavior displayed when access is restricted.

10. Testing

- Testing Strategy: Describe functional and access-control testing performed on the ACL.
- Test Cases: Test valid field values, invalid field values, authorized users, unauthorized users, and boundary/edge cases.
- Expected Results: Verify that the ACL consistently permits or denies access according to the configured rules.

11. Screenshots or Demo

- Configuration Screenshots: Include screenshots of the ACL configuration and script.
- Execution Screenshots: Include screenshots demonstrating successful access and restricted access.
- Demo: Provide a link to the project demonstration, if available.

12. Known Issues

- Document any known limitations, configuration dependencies, incorrect field-value scenarios, or access-control issues.
- Mention any cases requiring administrator intervention or additional configuration.

13. Future Enhancements

- Extend the ACL to support multiple field values and complex access conditions.
- Add reusable authorization functions and centralized access-control rules.
- Improve audit logging and reporting of allowed and denied record-access attempts.
- Add automated test coverage for ACL conditions and role-based access scenarios.